```python
import json
import anthropic
from config import (
    ANTHROPIC_API_KEY, SYMBOL_NAME,
    ATR_SL, ATR_TP1, ATR_TP2, ATR_TP3,
    TF_SCALP, TF_ENTRY, TF_TREND,
)

client = anthropic.Anthropic(api_key=ANTHROPIC_API_KEY)

_SYSTEM = f"""You are an expert {SYMBOL_NAME} scalping trader.
Analyze multi-timeframe data and return ONLY a JSON object — no markdown, no prea

JSON schema:
{{
  "signal":     "BUY" | "SELL" | "NO_TRADE",
  "confidence": 0-100,
  "entry":      number,
  "sl":         number,
  "tp1":        number,
  "tp2":        number,
  "tp3":        number,
  "rr":         "1:X",
  "timeframe":  "1m" | "5m",
  "setup":      "short name, e.g. EMA Cross + RSI Divergence",
  "bias":       "bullish" | "bearish" | "neutral",
  "reasoning":  "2-3 sentences max, actionable",
  "urgency":    "immediate" | "wait_for_confirmation" | "skip"
}}

Rules:
- Only signal BUY/SELL when confidence >= 65; otherwise NO_TRADE.
- Use ATR-based SL/TP distances provided.
- Align entry direction with 15m bias whenever possible.
- Keep reasoning concise and trader-focused."""


def generate_signal(analysis: dict, news: list[str] = None) -> dict:
    tf1  = analysis[TF_SCALP]
    tf5  = analysis[TF_ENTRY]
    tf15 = analysis[TF_TREND]
    atr  = tf5["atr"]
    price = tf1["price"]
```

```
    news_block = ""
    if news:
        news_block = "\n\nRecent Gold News:\n" + "\n".join(f"• {h}" for h in news

    prompt = f"""
{SYMBOL_NAME} Scalp Analysis — Current Price: {price}

── 15M (Trend Bias) ──────────────────
EMA Trend : {tf15['ema_trend']}  |  9/21/50: {tf15['ema9']} / {tf15['ema21']} / {
RSI       : {tf15['rsi']} ({tf15['rsi_state']})
MACD Hist : {tf15['macd_hist']}  |  Cross: {tf15['macd_cross']}
BB        : {tf15['bb_lower']} / {tf15['bb_mid']} / {tf15['bb_upper']}  |  Price:

── 5M (Entry Confirmation) ───────────
EMA Trend : {tf5['ema_trend']}  |  9/21: {tf5['ema9']} / {tf5['ema21']}
RSI       : {tf5['rsi']} ({tf5['rsi_state']})
Stoch K/D : {tf5['stoch_k']} / {tf5['stoch_d']} ({tf5['stoch_state']})
MACD Hist : {tf5['macd_hist']}  |  Cross: {tf5['macd_cross']}
BB        : {tf5['bb_lower']} / {tf5['bb_mid']} / {tf5['bb_upper']}  |  Price: {t
ATR       : {atr}  |  Vol Spike: {tf5['vol_spike']}

── 1M (Scalp Timing) ─────────────────
EMA 9/21  : {tf1['ema9']} / {tf1['ema21']}
RSI       : {tf1['rsi']}  |  MACD Cross: {tf1['macd_cross']}
Last 5 candles: {tf1['last_5']}

── ATR Risk Levels (ATR={atr}) ────────
SL dist   : {round(atr * ATR_SL,  2)}
TP1 dist  : {round(atr * ATR_TP1, 2)}
TP2 dist  : {round(atr * ATR_TP2, 2)}
TP3 dist  : {round(atr * ATR_TP3, 2)}
{news_block}

Return the JSON signal now:"""

    resp = client.messages.create(
        model="claude-sonnet-4-20250514",
        max_tokens=500,
        system=_SYSTEM,
        messages=[{"role": "user", "content": prompt}],
    )

    raw = resp.content[0].text.strip().replace("```json", "").replace("```", "").
    return json.loads(raw)
```